

## **PROGRAM 4 - Create a dockerized application and push it to GitHub using Personal Access Token (PAT).**

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### **Steps**

#### **1. Create project directory**

```
mkdir lab4 && cd lab4
```

#### **2. Create files**

- app.py
- requirements.txt
- Dockerfile

#### **3.Create Dockerfile**

Create Dockerfile:nano Dockerfile

```
FROM python:latest
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python", "app.py"]
```

#### **3. Build Docker image**

```
docker build -t program4 .
```

#### **4. Run Docker container**

```
docker run -p 5000:5000 program4
```

## 5. Verify output

Open browser: <http://localhost:5000>

## 6. Initialize Git

```
git init
```

```
git branch -M main
```

```
git add .
```

```
git commit -m "initial commit"
```

## 7. Push to GitHub using PAT

```
git remote add origin https://github.com/<username>/program4.git
```

```
git push -u origin main
```

## OUTPUT

Dockerized application created and pushed to GitHub successfully.

